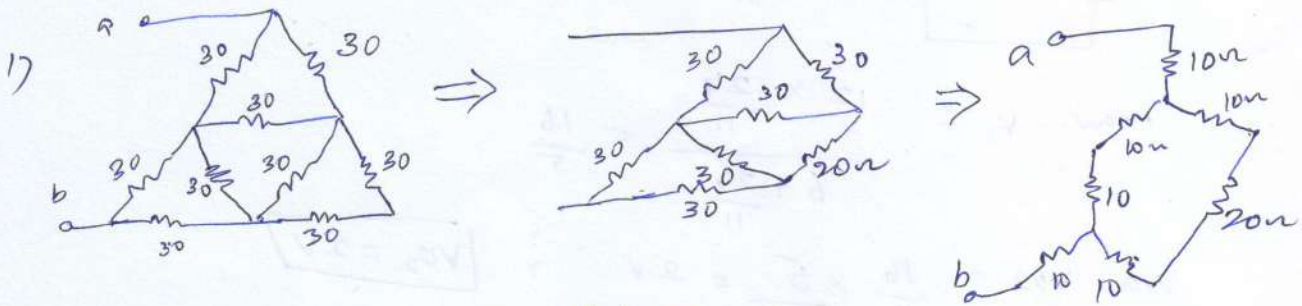


Slot: C1. key.

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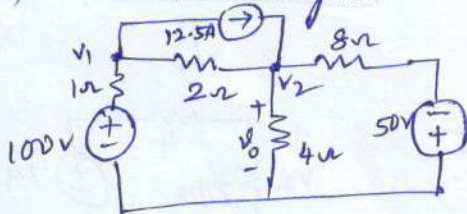
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$$R_{ab} = 10 + (10+10) \parallel (10+20+10) + 10$$

$$R_{ab} = 33.33 \Omega$$

2) Nodal analysis to find V_0 .



at node 1:

$$\frac{V_1 - 100}{1} + \frac{V_1 - V_2}{2} + 12.5 = 0$$

$$\therefore V_2 = V_0$$

$$3V_1 - V_0 = 200 - 25 = 175 \quad \text{--- (1)}$$

at node 2:

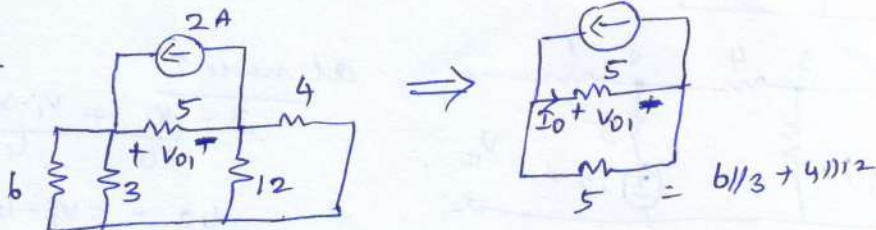
$$\frac{V_0 - V_1}{2} - 12.5 + \frac{V_0 - 0}{4} + \frac{V_0 - (-50)}{8} = 0$$

$$-4V_1 + 7V_0 = 50 \quad \text{--- (2)}$$

Solving for V_0 ; $V_0 = 50V$.

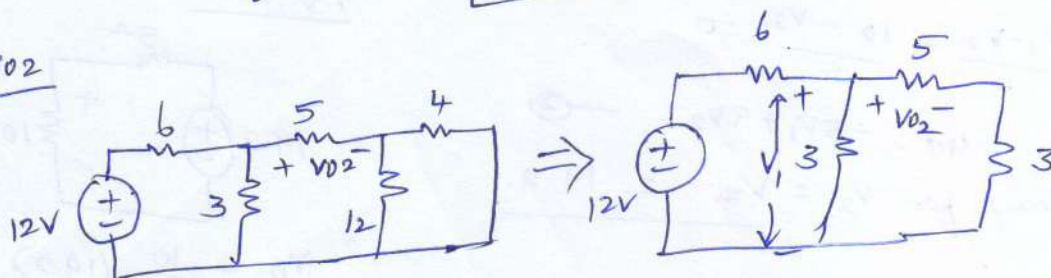
3) $V_{OT} = V_{O1} + V_{O2} + V_{O3}$.

V_{O1}



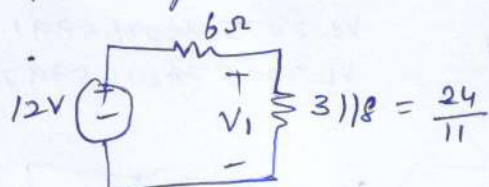
$$I_0 = 1A \therefore V_{O1} = 5V$$

V_{O2}



Finding V_1 will help to find V_{O2} .

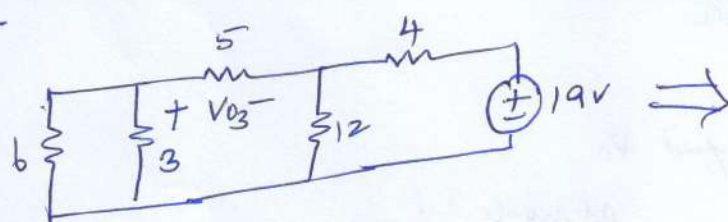
$\therefore V_1 = \text{drop across } 3\Omega \text{ common to 2 loops.}$



$$\text{Now. } V_1 = \frac{12 \times \frac{24}{11}}{6 + \frac{24}{11}} = \frac{16}{5}$$

$$\text{Now } V_{O2} = \frac{16}{5} \times \frac{5}{8} = 2V \quad ; \quad \boxed{V_{O2} = 2V}$$

V_{O3}

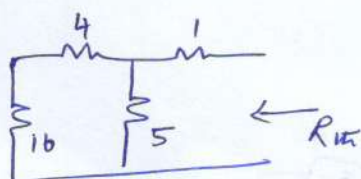


$$V_2 = \frac{19 \times \frac{7 \times 12}{19}}{\frac{7 \times 12}{19} + 4} = 9.975$$

$$\therefore V_{O3} = -7.125V$$

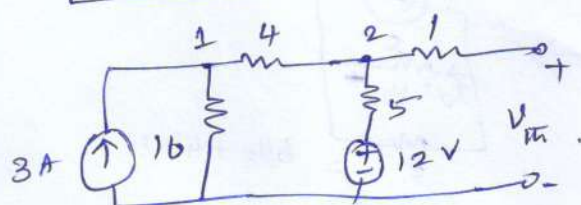
$$V_{TOT} = 5 + 2 - 7.125 = \underline{\underline{-125\text{mV.}}}$$

4. R_{th}



$$R_{th} = 1 + 5 || (4 + 16) = \underline{\underline{5\Omega}}$$

V_{th}



at node 1.

$$3 = \frac{V_1}{16} + \frac{V_1 - V_2}{4}$$

$$48 = 5V_1 - 4V_2 \quad \text{--- (1)}$$

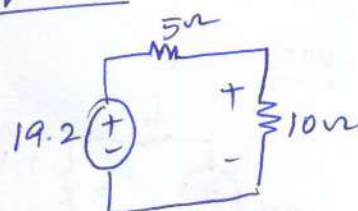
at node 2.

$$\frac{V_1 - V_2}{4} + \frac{12 - V_2}{5} = 0$$

$$48 = -5V_1 + 9V_2 \quad \text{--- (2)}$$

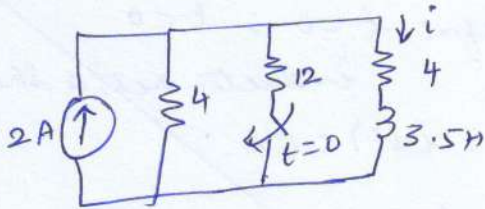
$$\text{solving for } V_2 = V_{th} = \underline{\underline{19.2}}$$

Eq. ckt



$$\therefore V_0 = \frac{10}{10 + 5} (19.2) = \underline{\underline{12.8V}}$$

5)



$$R_{eq} = 4 + (4 \parallel 12) = 7\Omega$$

$$i(0) = 1$$

$$i(\infty) = \frac{2 \times (4 \parallel 12)}{4 + (4 \parallel 12)} = \frac{6}{7}$$

$$i(t) = \frac{6}{7} + \left[1 - \frac{6}{7} \right] e^{-t/0.5}$$

$$\therefore i(t) = \frac{1}{7} (6 - e^{-2t}) A.$$

(3)

$$t < 0 \quad i(t) = \frac{2 \times 4}{4 + 4} = 1 A.$$

$$t > 0 \quad i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$\tau = \frac{L}{R_{eq}} = \frac{3.5}{7} = \frac{1}{2}$$